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Module 2 Introduction

Lesson 1: CNN for images

Lesson 2: Convolution, pooling and other layers

Lesson 3: CNN applications

Module 2 Summative Assessment

Graded Assignment: Module 2 Summative Assessment Submitted

Module 2 Summary

Your grade: 93.33%

Your latest: 93.33% • Your highest: 93.33% • To pass you need at least 80%. We keep your highest score.

Next item

Module 2 Summative Assessment

Review Learning Objectives

Instructions

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

Assignment details

Due
Attempts
Submitted
Nov 19, 3:00 AM IST

Retry

Your grade
To pass you need at least 80%. We keep your highest score.
93.33%

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1. What is the primary feature detected by the first convolution layer inAlexNet? 0 / 1 point

- High-level abstract features.
- High-frequency grayscale textures.
- Low-frequency color features.
- Specific object shapes.

Incorrect
Incorrect. The first layer typically detects edges and color features.

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2. Which application involves modifying the style of an image to mimic famous paintings? 1 / 1 point

- Feature extraction.
- Object detection.
- Semantic segmentation.
- Deep Style.

Correct
Correct!

3. What challenge is highlighted by the statement "Deep Neural Networks are Easily Fooled"? 1 / 1 point

- DNNs can overfit to the training data
- DNNs require large amounts of data to learn.
- DNNs can misclassify images that are slightly altered or contain noise.
- DNNs are computationally expensive to train.

Correct
Correct!

4. How do filters in a CNN operate? 1 / 1 point

- By fully connecting to all neurons in the previous layer.
- By scanning the entire image in one operation.
- By detecting specific features from the input data at various locations.
- By applying dropout to reduce overfitting.

Correct
Correct!

5. Which layerproduce probability for classification? 1 / 1 point

- Pooling
- Softmax
- Convolution
- ReLU

Correct
Correct!

6. Assuming an input shape is 1.3030, there are 36 3*3 filters for convolution layer. What is the output shape after convolution (stride = 1)? 1 / 1 point

- 36 * 28 * 28 (Valid convolution)
- 36 * 29 * 29
- 1 * 28 * 28
- 1 * 29 * 29

Correct
Correct!

7. Which layerchange the matrix to a vector? 1 / 1 point

- Pooling
- Flatten
- Convolution
- ReLU

Correct
Correct!

8. Assuming an input shape is 3.5050, what is the output shape after 2*2 max pooling layer? 1 / 1 point

- 3 * 25 * 25
- 1 * 25 * 25
- 3 * 24 * 24
- 1 * 24 * 24

Correct
Correct!

9. What is the primary goal of visualizing convolutional neural networks as discussed in the provided literature? 1 / 1 point

- To decrease model accuracy.
- To increase computational efficiency.
- To understand model predictions.
- To reduce the size of the model.

Correct
Correct!

10. How do CNNs handle patterns appearing in different regions of an image? 1 / 1 point

- By using different sets of parameters for each region.
- Through a fully connected feedforward network.
- By using the same set of parameters for detecting similar patterns.
- Ignoring the location of patterns within the image.

Correct
Correct!

11. How are filters applied in the CNN architecture for speech recognition? 1 / 1 point

- Filters move along the time axis only.
- Filters are static and do not move.
- Filters move in the frequency direction.
- Filters expand and contract dynamically based on the input.

Correct
Correct!

12. What is the purpose of subsampling in CNNs? 1 / 1 point

- To increase the size of the image.
- To change the object in the image.
- To reduce the spatial size of the image, leading to fewer parameters.
- To enhance the quality of the image.

Correct
Correct!

13. What determines the number of channels in the output after a convolution operation in a CNN? 1 / 1 point

- The size of the filters.
- The number of filters applied.
- The stride with which the filters are applied.
- The padding added to the input image.

Correct
Correct!

14. In the context of CNNs, what is the advantage of using shared weights and biases in convolutional filters? 1 / 1 point

- It allows the network to learn more complex patterns.
- It reduces the memory footprint and computational complexity.
- It increases the model's robustness to overfitting.
- It enables the model to process inputs of varying size.

Correct
Correct!

15. Which statement best describes the role of the Fully Connected (FC) layer in a CNN? 1 / 1 point

- It applies convolution operations across the entire input volume.
- It performs down-sampling to reduce the dimensionality of the input data.
- It combines features across the entire input volume to make predictions.
- It reduces the number of parameters by applying Max Pooling.

Correct
Correct!